

Towards Spatial Introspection and Experiential Propection: A Speculative Design Inquiry in Extended Reality

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ABSTRACT

Introspection involves examining one's thoughts, memories, feelings, and mental processes to gain deeper self-awareness, while propection refers to envisioning future scenarios and outcomes. In this speculative design inquiry, we present four design proposals that explore how XR combined with Artificial Intelligence (AI) and personal spatial memory (XRMemory) data collected from XR glasses could generate alternative perspectives on one's life to support introspection and propection. We conclude by examining the broader implications of our work, including ethical considerations and promising directions for future research in designing introspective and prospective XR experiences.

Index Terms: Speculative Design, Reflective Design, Spatial Memory, Embodied Memory, Introspection, Propection, Experiential Computing, Spatial Computing, Extended Reality

1 INTRODUCTION

Extended Reality (XR) technologies—encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)—are revolutionizing how we perceive and interact with digital information. Beyond entertainment and productivity, XR increasingly serves as a platform for deeper cognitive and emotional processes. In particular, immersive environments have the capacity to foster spatial introspection and experiential propection: two complementary phenomena that invite users to reflect on their present selves while envisioning alternate futures. Spatial introspection uses the tangible, embodied qualities of XR to anchor users' memories, emotions, and beliefs in three-dimensional space. This setting facilitates novel forms of self-awareness, where individuals can explore and even manipulate their inner experiences in real-time. Conversely, experiential propection leverages XR's interactive simulations to let individuals project themselves into possible futures—be it personal growth paths, career scenarios, or communal visions—thus linking present actions to aspirational outcomes.

Major technology companies such as Apple, Google, and Meta are pushing the boundaries of Extended Reality (XR), envisioning a future in which XR smart glasses become as commonplace as smartphones are today. This marks a profound shift in the near future: XR smart glasses will be worn throughout the day, capturing everything users see and hear, and continuously functioning as external recorders and manipulators of “spatial memory.”

This paper argues that XR's ability to merge presence, immersion, and adaptive feedback uniquely positions it as a medium to deepen self-reflection and forward-thinking utilizing spatial memory. By examining how design strategies—such as embodied metaphors, adaptive AI, and privacy-conscious data handling—can support these processes, we illuminate a largely untapped potential within the realm of XR. The following sections detail the theoret-

ical foundations, propose design patterns and use cases, and identify ethical considerations that must guide any further development. Our overarching aim is to highlight a new paradigm where technological immersion transcends entertainment, prompting transformative insights and motivated, future-oriented behaviors in diverse domains.

2 RELATED WORK

2.1 XR and Self-reflection

Early explorations of virtual reality (VR) and augmented reality (AR) focused on heightened presence, realism, and user engagement [20]. More recent work shifts toward therapeutic and self-reflective applications, demonstrating how immersion can foster deeper emotional awareness [1]. For instance, VR-based exposure therapy has shown promise in treating phobias and post-traumatic stress by situating users in controlled, yet evocative, virtual contexts [13]. Parallel research on memory palaces [11] and embodied cognition [22] highlights how spatial layouts in VR can aid knowledge retention and introspective processes. These studies suggest that virtual spaces can serve as “containers” for personal reflection, anchoring abstract emotional or cognitive states in immersive environments.

2.2 Embodied Memory

The concept of embodied memory has garnered attention across disciplines due to its relevance in understanding how memory is deeply intertwined with bodily experiences and actions. Embodied memory emphasizes that recollection is not purely cognitive but fundamentally linked to sensory, motor, and contextual interactions.

In cognitive science, theories of embodied cognition highlight the role of the body in shaping memory processes [3]. Research has demonstrated that memory retrieval is often facilitated by reenacting sensory or motor states associated with the original experience [6]. These findings align with phenomenological approaches, such as those of Merleau-Ponty, which underscore the inseparability of bodily perception and memory [14].

2.3 Embodied Memory and Mixed Reality

The integration of Mixed Reality (MR) technologies has opened new avenues for exploring embodied memory. MR environments, which combine real and virtual elements, provide immersive platforms where sensory and motor experiences can be simulated or augmented to evoke memory. For example, virtual environments have been used to recreate historical or personal scenes, allowing users to engage with memories in a multisensory manner.

Studies on MR's impact on memory have shown its potential to deepen engagement and enhance recall by leveraging embodied interactions. Users in MR environments can physically navigate spaces, interact with objects, and experience sensory cues that trigger memory processes. The sense for both presence and seminar memory recall in the “real” condition were statistically higher [12].

At the same time, critical perspectives raise questions about authenticity and the potential for altered or reconstructed memories in MR experiences. As digital representations simulate or reinterpret reality, they may influence how memories are perceived, stored, or

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even redefined [10]. These challenges underscore the need for further exploration of the ethical and psychological implications of embodied memory in MR.

2.4 Reflective Design in HCI

The concept of Reflective Design [18] promotes systems that encourage users to pause, question assumptions, and examine internal beliefs. In human-computer interaction (HCI), reflective tools often combine visualizations of personal data—like mood tracking or digital journals—with interactive prompts that foster self-awareness [19]. Similar trends in persuasive technology [7] emphasize the power of well-structured feedback loops for behavior change. These methods intersect with introspection by urging users to recognize patterns and motivations underlying their actions.

2.5 Prospecction and Future-Thinking

Prospecction—envisioning possible futures—has long been studied in psychology [17]. Traditional approaches involve scenario planning, goal-setting, and visualization exercises; when brought into XR, these become experiential rather than purely mental. Preliminary work in simulation-based learning [5] shows that immersive role-play can enhance empathy, strategic thinking, and motivational commitment to long-term objectives. Beyond individual applications, speculative design [2, 8] and design fiction [4] encourage collective imagining of alternative futures. By leveraging XR's capacity for real-time, dynamic interaction, users can “live out” prospective scenarios, testing outcomes that might otherwise remain abstract or inaccessible.

2.6 Integrating Spatial Introspection and Experiential Prospecction

Recent interdisciplinary research suggests that XR environments integrating introspective cues (e.g., mood-driven world transformations) with prospective simulations (e.g., branching life narratives) can yield powerful experiences [15]. Such designs blur the boundary between present self-reflection and future self-projection. However, challenges remain regarding personalization, emotional safety, and ethical data handling. As XR systems grow more context-aware, the integration of physiological sensors, AI-driven adaptivity, and multisensory feedback opens new frontiers for designing immersive, user-centric experiences that guide both inward exploration and outward aspiration.

3 CONCEPTUAL FRAMEWORK

This paper presents a conceptual framework that integrates two core XR design principles: spatial introspection and experiential prospecction. These principles underscore the transformative potential of immersive technologies for self-reflection and future envisioning. The framework is structured around the following key components:

Dis-Embodied Introspection: XR enables users to engage in introspection from a detached, third-person perspective. Tools like ThirdEye reconstruct past experiences and present them from an elevated, bystander viewpoint. By stepping outside their own embodiment, users can observe their behaviors, emotions, and interactions with greater clarity, free from the constraints of immediate biases or emotions. This detached perspective fosters self-awareness and helps users gain insights into their actions and decisions.

Embodiment Shifts Across Perspectives and Time: XR memory enables users to shift their embodiment into different participants or vantage points within a shared experience. By synchronizing spatial and temporal data, users can explore how events unfolded from multiple perspectives and even across different time frames. This capability allows users to revisit key moments through diverse lenses, promoting a deeper understanding of collective dynamics and individual contributions.

Experiential Prospecction: XR projects users into immersive, sensory-rich simulations of plausible future scenarios, turning abstract aspirations into vivid, participatory experiences. Tools like DeJaVision utilize self-suggestion to create “memories from the future,” fostering intentionality and alignment between present actions and desired outcomes. These simulations encourage users to rehearse key decisions or explore alternative futures in a safe, controlled environment.

Integration of Reflection and Envisioning: Spatial introspection and experiential prospecction form a continuous cycle, where insights gained from reflecting on the past inform personalized future scenarios. Conversely, prospective simulations inspire deeper introspection and self-refinement. This interplay supports user progression through a coherent narrative of self-awareness and forward-thinking.

4 SPATIAL INTROSPECTION AND EXPERIENTIAL PROSPECCTION

Augmented Reality (AR) glasses will serve as tools for recording daily life, transforming personal experiences into design resources. In this paradigm, input signals are no longer solely processed through the human body—remembered, forgotten, and reorganized into memories—but are also intercepted and captured by Mixed Reality (MR) devices such as glasses or headsets. As the adoption of MR technologies becomes increasingly widespread, it is imperative to move beyond a focus on hardware and critically examine the potentials and challenges associated with this new medium for memory retention.

A central question arises: how can this vast repository of data be harnessed to benefit both individuals and communities? To address this, we propose several prototypes that leverage XR memory as a means for spatial introspection—allowing users to reflect on past experiences within their physical environments—and experiential prospecction, enabling individuals to imagine and prepare for future scenarios.

4.1 ThirdEye: Self-Reflection from Different Perspective

Recent works like Zenbu Koko [21] and Designing a Virtual Death VR Experience [9] have demonstrated the reconstruction and display of participants' bodies in virtual environments, allowing them to view their reconstructed forms from an elevated perspective. As spatial XR memory becomes accessible to everyone, these memories will provide people with a third-eye view to observe themselves as bystanders. This perspective can help people in various situations—from posture correction to reexamining life's conflicts and unspoken words—enabling users to step back from their biases and emotions that constrain them, to gain clearer insights into their behaviors and decisions.

Moreover, integrating therapeutic techniques like Two-Chair Therapy into ThirdEye enables users to engage in simulated dialogues with themselves or representations of others. For example, users can alternate between their reconstructed self and a virtual counterpart to explore unresolved conflicts, practice empathy, or achieve emotional closure.

4.2 AlterPast: Rehearsal the “What if” Moments

AlterPast is an interactive tool designed to enable users to “rehearse” their past experiences while fostering performative healing and empowering them to play a better self. In daily life, it is common to reflect on pivotal moments and wonder, “*What if I had done that differently?*” or “*What if I had said this instead?*” People also often imagine how others might act in similar situations, adopting a role-playing mindset such as, “*If my teacher were here, what would they do?*”

By leveraging AI and data collected from XR memories, AlterPast can generate AI-driven agents modeled after people in the user's close relationships, such as a spouse, child, parent, or colleague. These agents, represented as holographic avatars, simulate realistic behaviors and responses based on past interactions and contextual data. Within a recreated "past scene," users can engage interactively with these avatars, experimenting with different reactions, tones, or choices in conversations or activities. This allows them to explore alternative outcomes, fostering improved understanding, conflict resolution, and enhanced collaboration.

Importantly, AlterPast is not intended to encourage regret over past actions but to serve as a practice and refinement tool for real-life communication skills. By revisiting key moments and experimenting with alternate approaches, users can gain deeper insights into their interpersonal dynamics and cultivate more empathetic, constructive, and effective interactions in future scenarios.

Applications for AlterPast range from personal development to professional training, offering individuals a unique way to reflect, learn, and grow through immersive experiential rehearsal.

This embodied introspection brings the following opportunities:

Performative Healing: AlterPast serves as a therapeutic space where users can revisit unresolved moments from their past, not to dwell on regrets, but to rehearse healthier responses and emotional resolutions. By reenacting critical interactions in a controlled, immersive environment, users can gain clarity, process lingering emotions, and transform their perspective on difficult situations.

Play a Better Self: AlterPast empowers users to step into the role of their "ideal self" in challenging scenarios. Whether practicing assertiveness, demonstrating empathy, or navigating conflict, users can rehearse new behaviors and approaches that align with their aspirations for personal growth. This performative aspect helps bridge the gap between who they are and who they wish to become, creating a pathway for real-world transformation.

4.3 SpatialHistory: Community-based Collective Memory Explorer

As XR memory becomes widespread, people will be able to collect and combine memories from multiple individuals experiencing the same event, creating a panoramic XR reconstruction. Similar to multiplayer games, these memories can be synchronized spatially and temporally where they overlap, then merged into a cohesive whole. This would allow us to reconstruct diverse perspectives—from the separated tours of the bride's team and the groom's team at a traditional wedding celebration, various attractions at a holiday fair, or simultaneous conference sessions occurring between satellite venues and the main stage. It might bring:

Memory Contribution Frameworks: The collective memory requires a system that allow people to voluntarily contribute XR data, with tools to anonymize sensitive details or highlight key moments, and reconstruct it spatially and temporally.

Shared, Shifting Embodied Perspectives: By collecting memories (video, audio, and spatial data) from different participants in an event, the spatial memory of a community is recorded and become possible for revisits. Participants can revisit and even shift their embodiment in this collective memory, and re-live that important day in their life in another way.

Time-Lapse and Rewind: Participants could navigate through the collective memory with unprecedented temporal control, moving freely through the event's timeline to witness moments unfold from multiple perspectives. They could fast-forward through hours of experiences in seconds, pause at significant moments to examine details from different vantage points, or rewind to revisit particularly meaningful interactions. This temporal flexibility would allow users to discover previously unnoticed connections and patterns, gaining deeper insights into how events unfolded across different locations and perspectives simultaneously.

New Form of Storytelling: Collective XR memories could fundamentally transform traditional narrative structures by enabling rich, multi-layered storytelling experiences that weave together diverse perspectives and viewpoints. Unlike linear narratives, these immersive stories would allow audiences to explore events through multiple simultaneous viewpoints, creating a more nuanced and comprehensive understanding of shared experiences. This novel approach to storytelling could break down the barriers between narrator and audience, allowing for more democratic and participatory forms of narrative creation where multiple voices and interpretations can coexist and interact within the same spatial-temporal framework.

4.4 DejaVision: Experiential Manifestation

DejaVision is an AI-driven tool that leverages experiential manifestation to create vivid, immersive "memories" of potential future scenarios. By generating highly personalized, realistic simulations of future moments, it creates a sense of *déjà vu*—the eerie familiarity of experiencing something that has seemingly already happened. This tool combines elements of self-suggestion and memory-building, aiming to influence decision-making and inspire personal growth.

DejaVision blurs the boundary between imagination and reality, embedding these AI-generated scenarios in the user's mind as "memories from the future." Through this process, it fosters intentional action and aligns personal or communal choices with desired outcomes.

AI-Generated Future Scenarios as memory: DejaVision uses AI to model future environments and interactions based on a user's past behaviors, XR memories, and goals. These scenarios are rendered as immersive XR experiences, allowing users to "visit" their potential future in vivid, multisensory detail. For example, a user might experience delivering a successful keynote speech years before these events occur.

Self-Suggestion for Behavioral Alignment: By immersing users in positive, aspirational futures, DejaVision leverages the psychological power of *self-suggestion*. The brain perceives these scenarios as potential realities, increasing motivation to act in ways that make them come true.

For example, experiencing the sensation of being a confident interviewee in a high-stakes situation can help users embody that confidence in that scenario.

5 DISCUSSION

This paper demonstrates how XR technologies can transform self-awareness and forward-thinking through the integration of spatial introspection and experiential prospection. The four prototypes presented—ThirdEye, AlterPast, SpatialHistory, and DejaVision—highlight the potential of XR to reframe how individuals reflect on the past, envision the future, and interact with others.

5.1 Discussion on the Design Prototypes

5.1.1 Spatial Memory as a Key Resource for Personal Growth

Despite the human brain's remarkable capabilities, it remains vulnerable to what Schacter calls the "seven sins of memory," grouped into three categories: forgetting (e.g., absent-mindedness), distortion (e.g., misattribution), and persistence (e.g., intrusive memories) [16]. These limitations highlight the potential value of a robust external memory system—one that continuously captures and archives our lived experiences.

XR smartglasses, worn throughout the day, can record high-fidelity RGB video, audio, and depth data from the wearer's perspective. This external, persistent collection of "spatial memory" has never been more feasible, thanks to declining storage costs. Indeed, smartglasses represent a paradigm shift: they do not merely

“augment” reality; they record it, forming a rich repository of personal data. This repository can serve as a powerful resource for self-reflection, learning, and broader personal development. Through XR and AI, individuals might revisit meaningful life episodes, gain greater awareness of their present habits, and even explore plausible futures—thus bridging limitations in human recall and reshaping how people engage in introspection and prospection.

5.1.2 Expanding the Design Space of Introspection

Introspection has traditionally focused on conscious, subjective reflections—through journaling, meditation, cognitive-behavioral techniques, and other psychological practices. Most of these methods rely heavily on memory recall, whether from immediate experiences or from deeper autobiographical narratives. As HCI research has increasingly recognized the importance of reflection for personal well-being, spatial memory adds a transformative layer.

“Spatial introspection” can be defined as a reflective process that uses immersive, first-person data—captured via XR devices—to revisit, relive, and even reconfigure personal past experiences. Instead of merely recounting events abstractly, spatial introspection allows users to embody their recollections.

5.2 Limitation

While the presented framework and prototypes demonstrate the transformative potential of XR technologies, several limitations must be acknowledged:

1. **Accuracy of XR Memory Reconstruction:** The reliability of XR memories depends on the quality and completeness of the data collected. Incomplete or biased recordings can lead to distorted representations of past events, which may hinder effective introspection or lead to misinterpretations during simulations.
2. **Ethical and Privacy Concerns:** The use of personal and collective data to reconstruct memories or simulate scenarios raises significant privacy concerns. Ensuring that users consent to data collection and understand how their information will be used is essential, yet achieving transparency and security in such systems remains challenging.
3. **Memory Manipulation and False Memories:**
XR systems can alter how individuals recall past events, potentially implanting false memories if AI-generated reconstructions or simulations deviate from actual lived experiences. Users may conflate AI-generated scenarios with reality, especially if they frequently relive or tinker with them in XR. This conflation raises profound questions about authenticity, identity, and the reliability of personal memory.
4. **AI Hallucination and Psychological Risks:**

Machine learning models might introduce inaccuracies or “hallucinations,” creating plausible yet entirely fabricated details in reconstructed or forecasted scenarios. When woven into introspection or prospection practices, these hallucinations can distort users’ sense of self, undermine trust in the technology, and potentially lead to maladaptive behaviors.

6 CONCLUSION

Spatial introspection and experiential prospection offer a powerful paradigm for exploring how Extended Reality (XR) can facilitate deeper self-awareness and forward-thinking. By integrating introspective reflection with immersive simulations of potential futures, XR technologies transcend traditional screen-based paradigms, inviting users to “step inside” their memories, emotions,

and aspirations. This dual focus not only unlocks novel design possibilities—such as adaptive, embodied interactions and contextually aware environments—but also carries significant implications for personal well-being, learning, and organizational innovation.

However, realizing this vision demands careful attention to ethics, data privacy, and psychological safety. As user experiences become more embodied and data-rich, transparent design methods, user agency, and culturally sensitive customization become critical. Interdisciplinary collaboration across HCI, cognitive science, design, and ethics will be vital to ensure that these immersive experiences remain both profoundly engaging and responsibly crafted. In sum, the potential of spatial introspection and experiential prospection highlights XR’s promise as a transformative medium, fostering meaningful self-exploration and future-oriented growth.

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